

RESTORATIVE EFFECTS OF NATURAL ENVIRONMENT EXPERIENCES

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ABSTRACT: The utility of different theoretical models of restorative experience was explored in a quasi-experimental field study and a true experiment. The former included wilderness backpacking and nonwilderness vacation conditions, as well as a control condition in which participants continued with their daily routines. The latter had urban environment, natural environment, and passive relaxation conditions. Multimethod assessments of restoration consisted of self-reports of affective states, cognitive performance, and, in the latter study, physiological measures. Convergent self-report and performance results obtained in both studies offer evidence of greater restorative effects arising from experiences in nature. Implications for theory, methodology, and design are discussed.

"Climb the mountains and get their good tidings. Nature's peace will flow into you as sunshine flows into trees. The winds blow their own freshness into you and the storms their energy, while cares drop off like autumn leaves." John Muir

Poets, writers, philosophers, and artists have long held that natural settings are good for body, mind, and soul. More recently, researchers have begun to examine this belief in the

salutary effects of nature. Unfortunately, too much of this research has relied on atheoretical and poorly designed program evaluations. Studies have suffered from self-selection of subjects into samples and an absence of either comparison groups or random assignment when comparisons have been included. They have also been characterized by inadequate follow-up procedures and overreliance on self-report measures. Nonetheless, they have shown a remarkable degree of convergence. In almost all cases no harmful effects have been uncovered, while either marginal or significant gains have been detected in outcomes such as competence, positive affect, and self-esteem. Reviews of this literature are provided by Driver, Nash, and Haas (1987), Ewert (1988), Gibson (1979), and Levitt (1989).

The research programs of two groups of scientists have uncovered similarly consistent findings using more rigorous research designs built upon theoretical frameworks. In the earlier of these (summarized in R. Kaplan & S. Kaplan, 1989), studies conducted over more than a decade measured emotional and cognitive changes in participants in an outdoor challenge program. The program involved instruction in outdoor

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survival skills within the context of an approximately two-week-long wilderness outing. Before-and-after comparisons as well as cross-sectional analyses with control groups indicated gains in confidence and self-sufficiency, more realistic self-evaluations, and enhanced outdoor survival skills. Early studies in this series found that these changes in feelings and skills persisted for over five months beyond the treatment (R. Kaplan, 1974, 1977). Qualitative analyses of participants' journal entries enriched and corroborated the quantitative results (S. Kaplan & Talbot, 1983).

The Kaplans' program of research contributed to a theory of restorative environments in which the reduction of mental fatigue is central to restoration (S. Kaplan & Talbot, 1983; R. Kaplan & S. Kaplan, 1989). It notes that many daily activities in contemporary society demand directed attention. Effort is required to gate out competing stimuli or thoughts so that one can pay attention. In the face of continual demand over time, this inhibitory control can be lost and the capacity for directed attention can become depleted. The resultant mental fatigue can be manifest in, for example, reduced effectiveness in functioning and reduced tolerance for frustration (Cohen & Spacapan, 1978; S. Kaplan, 1987; see also Cohen, 1978).

In an early phase of theory development, Kaplan (1978) built on James' (1892) distinction between voluntary and involuntary forms of attention in positing a way in which the capacity for directed attention might be restored. In contrast to directed or voluntary attention, little or no effort is required by an involuntary form of attention that Kaplan calls fascination. It comes into play when, out of interest, curiosity, and the like, certain objects (e.g., animals) and processes (e.g., exploration) capture and hold the attention. As involuntary attention is engaged, demands on a depleted voluntary attentional capacity are diminished. Its restoration then becomes possible. Thus one key aspect of restorative settings is their potential for eliciting fascination.

In addition to fascination, restorative settings should promote some sense of being away, extent, and compatibility (S. Kaplan & Talbot, 1983; R. Kaplan & S. Kaplan, 1989). Being away refers

to change in the location and/or activities of daily life. The importance of being away, or temporary escape from one's usual setting or situation, has been well documented in studies of the motivations of wilderness users and other outdoor recreationists (Driver & Knopf, 1976; Knopf, 1983, 1987). Extent refers to the sense that the immediate setting is part of some larger place or whole. It derives from a sense of the relatedness of immediately perceived setting elements and from cues that suggest that further information is available upon exploration (e.g., a bend in a path, a partially occluded clearing). Finally, compatibility refers to the fit or congruence between opportunities afforded by the setting, the constraints on behavior it imposes, and the individual's purposes and inclinations (S. Kaplan, 1983).

The second major program of research on restorative effects of natural environments has been conducted by Ulrich and colleagues (e.g., Ulrich, 1979, 1984; Ulrich & Simons, 1986). Ulrich's (1983) model of restoration, while in some ways similar to the Kaplans' model, differs from it in important ways. First, Ulrich's model assumes that the initial response to the environment is affective rather than cognitive. Working from Zajonc's (1980) concept of *preferenda*, he sees gross structural properties of the visual array (e.g., complexity, focality) eliciting deep-seated and nearly automatic responses in the absence of extensive information processing.

Second, Ulrich's model is more concerned with emotional, mental, and physiological components of response to taxing or threatening stimuli than with attention-based deficits arising from everyday activity. Somewhat similar views of nature as a stress-reducing setting are also salient in the wilderness and outdoor recreation literature (Knopf, 1987).

Third, in Ulrich's view restoration derives more from reduction of arousal than from replenishment of attentional capacity, which is instead viewed as a corollary of arousal reduction. Specifically, Ulrich suggests that nature has a calming effect because it is a nontaxing stimulus that elicits positively toned emotional states and blocks negatively toned feelings. Associ-

ated with these immediate affective responses are a reduction in neurophysiological arousal and a continued interest in the stimulus. These responses are thought to occur because natural settings continue to be associated with more positive affect through predispositions to respond positively to surroundings conducive to well-being or survival.

In keeping with the above assumptions, much of Ulrich's research has looked at emotional and physiological recovery from stress. Laboratory studies have examined restoration from naturally induced stress (from a college examination; Ulrich, 1979) and from stress induced in the laboratory setting (with an industrial accident film; Ulrich & Simons, 1986). In the former study, subjects who viewed slides of nature scenes reported greater posttest positive affect than subjects who viewed slides of urban settings. Likewise, those in the latter study who watched photographic simulations of natural settings showed faster physiological recovery than those who saw simulated urban settings. The physiological measures used were skin conductance, muscle tension, and pulse transit time. Ulrich (1984) also showed that patients who viewed natural elements through their hospital window had faster and more painfree recovery from surgery than patients whose view was of a brick wall.

OVERVIEW OF STUDIES

The present research attempts to build on the programs of research discussed above while at the same time addressing some of the more serious methodological problems that have plagued research on the benefits of natural environment experiences. Reports on two field studies are presented. The studies analyze the impacts of experiences of varying duration in settings thought to be of varying restorativeness. Both studies include comparisons involving alternative options for restorative experiences (e.g., vacations, relaxation). To help ensure unconfounded comparisons, both studies employ sound strate-

gies for control of self-selection (e.g., use of backpacking participants only, random assignment). Finally, to further increase confidence in the validity of results, both studies use a multi-method measurement strategy that includes self-report measures of emotional restoration and a behavioral measure of mental restoration. The second study includes physiological measures as well. Studies 1 and 2 are presented separately, followed by a general discussion.

STUDY 1

The first study examined the effects of an extended wilderness backpacking trip in comparison to the effects of other vacations and no vacation. Because random assignment was not an option, given the nature of the different conditions, self-selection of individuals into groups presented a problem. Individuals who would volunteer for a wilderness trip might differ significantly from others who would not choose to participate in such activities. To address this problem, only experienced backpackers were used as subjects. Level of physical fitness was also assessed and controlled. Many evaluations of outdoor programs may have confounded differences in fitness with effects of the nature experience per se.

Method

Design. A nonequivalent control group, quasi-experimental design was employed. Comparisons were made of experienced, physically fit backpackers who went on either a wilderness backpack ($n = 25$) or nonwilderness vacation ($n = 18$), or who remained at home engaged in their normal daily routines ($n = 25$). Both wilderness and nonwilderness vacations were from 4 to 7 days in duration. Pre- and posttest measures of affect and cognitive performance were taken. Affect was measured once again 21 days after the posttest.

Participants. All participants had backpacked on at least three overnight trips prior to the study. All participants engaged in

regular aerobic exercise and reported being in good physical condition. Members of the wilderness backpacking group were recruited at trailheads in California's Sierra Nevada mountains. Control groups were drawn from several sources, including a local outing club, members of the local Sierra Club, and from referrals provided by initial contacts. More detailed information on participants is provided in Table 1.

Control variables. Data on personal background, physical fitness, and attitudes toward wilderness were collected. Background data was gathered with a questionnaire that asked about age, gender, occupation, education, current residence, membership in conservation organizations, backpacking experience, and, as appropriate, anticipation of the upcoming vacation. Measures of physical fitness included a 5-point rating scale of current fitness and a question about frequency of aerobic workouts. Attitudes toward wilderness were measured with the Wilderness-Urbanism scale (Hendee, Catton, Marlow, & Brockman, 1968), which consists of 30 items describing features and activities associated with recreation in outdoor surroundings. Respondents indicate their extent of like or dislike for each item. Information on characteristics of the vacations (e.g., size of group, duration) was also collected.

Dependent variables. Self-reports of affective state included the Zuckerman Inventory of Personal Reactions (ZIPERS; Zuckerman, 1977) and the Overall Happiness scale (OHS; Campbell, Converse, & Rodgers, 1976). The ZIPERS consists of 12, 5-point scales that measure emotional states at the time the test is administered. Factors include fear arousal, positive affect, anger and aggression, sadness, and attentiveness and concentration. Extensive psychometric work indicates excellent reliability and validity for the scale (Zuckerman, 1977). The OHS is a magnitude estimate scale presented as a thermometer ranging from 0 (not at all happy) to 100 (very happy). This measure and a similar one assessing overall life satisfaction were taken from the American Quality of Life survey (Campbell, Converse, & Rodgers, 1976).

The behavioral measure of mental restoration was a proof-reading task adapted from Glass and Singer's (1972) research on stress aftereffects. Participants were asked to find misspellings, typographical errors, and grammatical errors in a five-page passage from an urban sociology text. Participants were encouraged to work quickly but accurately. The dependent variable was the percentage of errors discovered among the total number of errors that could have been discovered at the point in the text subjects reached after 10 minutes. Half the subjects had one passage of text to proofread before the treatment and half had a different passage. Following the treatment each group had the passage they had not seen before. The two passages were equivalent in number and type of errors. Pilot testing indicated that average performance scores for the two passages were similar.¹

Procedure. Backpackers approached at trailheads were asked to participate in the study if they passed initial screening with respect to length and difficulty of their planned trip and their prior backpacking experience. After pretesting eligible individuals, arrangements were made to meet with the participants at their exit points, where posttesting took place. Quiet areas away from parking lots and trails were chosen as test settings so that participants would not be distracted during testing.

Participants in the nonwilderness vacation group were pretested before they embarked on their trips and posttested within hours after their return. Typical nonwilderness vacations included car touring, visits to family/friends, and sightseeing trips. Members of the nonvacation control group were tested while observing a 4 to 7 day time lapse between pre- and posttesting. During the test period participants in this group carried on with everyday activities in the Los Angeles-Orange County metropolitan area of Southern California.

The sequence in which dependent measures were administered was held constant over time as much as possible. At pretesting, participants were instructed to complete the ZIPERS first, the OHS second, then the proofreading task, and, finally, the Wilderness-Urbanism scale. The sequence at posttesting

was as follows: ZIPERS, OHS, the proofreading task, and the background information questionnaire. The follow-up was administered through the mail. Participants were instructed to first complete the ZIPERS and then the OHS. The proofreading measure was not repeated at the follow up because of the difficulty of administering the task indirectly. For more details on Study 1 methods, see Mang (1984).

Results and Discussion

Given the quasi-experimental design, an important issue is the equivalence of the three groups in the absence of the treatment. As shown in Table 1, participants in the three groups were similar in many respects in background characteristics, backpacking experience, and wilderness attitudes. Trip characteristics were also similar. The selection criteria, pre-post design, and use of statistical controls help minimize the threat of self-selection as an alternative explanation for restorative effects. As noted earlier, those prior studies that did include control groups have typically compared participants in various outdoor programs to nonparticipants. Unfortunately, plausible self-selection alternative hypotheses cannot be ruled out for such studies.

The groups were similar in most respects as shown in Table 1. Covariance analysis was used to control for those variables that distinguished between groups and were correlated with outcome measures. Note that demographic characteristics of the sample are comparable to those described in other studies (e.g., Hendee et al., 1968).

Pre- and posttest means for the three groups are shown in Table 2. These data suggest and statistical analyses confirm that the groups did not differ in pretest measures with the exception of the ZIPERS positive affect factor, which was higher for the backpacking group, $F(2, 65) = 5.60, p < .001$.

For analyses in Study 1 the significance level was set at $p = .10$. This was done for several reasons. The study is exploratory and accordingly the need to uncover possible data trends

TABLE 1
Background Variables for Participants in Study 1

<i>Variable</i>	<i>Wilderness</i>	<i>Vacation</i>	<i>No Vacation</i>
Age	35.9	29.2	31.6
Gender			
Percentage male	64	56	64
Education			
Percentage some college	17	33	8
Percentage college graduate	50	17	32
Percentage post graduate work	29	50	56
Place of birth			
Percentage foreign*	4	28	4
Current residence			
Percentage urban	96	95	100
Years in residence*	11	4	4
Personal perceptions of physical fitness	2.16 ^a	2.39	2.12
Fitness schedule			
Percentage ≥ 3 times per week	64	55	68
Number in group	2.80	7.44	1.00
Age at first backpacking trip	18.4	17.1	17.3
Number of weeks since last trip	NA	26.9	29.0
Frequency of backpacking			
percentage 1-2 trips per year	64	61	44
Positive anticipation of trip*	1.32 ^a	1.78	NA
Days between pre- and posttest*	5.04	6.44	6.04
Wildernism-urbanism score			
(Hendee et al., 1968)	78 ^b	70	74

NOTE: NA = Not applicable.

a. Items rated on a scale where 1 = very much and 5 = not at all.

b. Scores are on a 90-point scale where low values reflect a high degree of urbanism and high values reflect strong positive views toward wilderness.

*Groups different at $p < .05$.

worthy of follow up is greater than the need to protect against false positive decisions (Myers, 1972). Also, the data come from a field study with minimal control over extraneous variation, which tends to decrease statistical power.

TABLE 2
Means for Pretest, Posttest, and Follow-Up Measures of Emotion

Dependent Variable	Wilderness Backpacking			Nonwilderness Vacation			No Vacation Control		
	Pre	Post	Follow-up	Pre	Post	Follow-up	Pre	Post	Follow-up
Fear	1.52	1.53	1.89	1.40	1.16	2.27	1.45	1.30	1.72
Anger and aggression	1.00	1.88	1.04	1.22	2.16	1.16	1.00	1.52	1.12
Sadness	1.08	1.08	1.40	1.16	1.38	1.50	1.24	1.44	1.28
Positive affect	3.66	3.84	3.48 ^a	3.01	3.33	3.77	2.91	3.12	3.36
Attentiveness	3.28	2.88	3.36	3.16	2.50	3.11	3.72	2.28	3.44
Overall happiness	76.0	59.2	87.6 ^b	67.2	67.8	78.9	72.7	73.2	78.4
Overall satisfaction	70.0	68.0	88.0	73.9	66.7	79.4	74.0	73.2	78.0

NOTE: The Fear through Attentiveness measures were scaled from 1 = low to 5 = high. The Overall happiness measure was scored from 0 = Not at all to 100 = Very happy.

a. The backpacking groups *pretest* mean is significantly greater than the pretest means of the other two groups.

b. The backpacking groups *posttest* mean is significantly greater than the posttest means of the other two groups.

TABLE 3
Group Means for the Pretest and Posttest Measures of Attentional Capacity

<i>Dependent Variable</i>	<i>Wilderness Backpacking</i>		<i>Nonwilderness Vacation</i>		<i>No Vacation Control</i>	
	<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>	<i>Pre</i>	<i>Post</i>
Proofreading performance (percentage errors detected) ^a	51.12	53.92	56.77	53.78	60.16	56.52

a. The Time $\times$ Group interaction is statistically significant at $p = .09$, and with the two control groups collapsed at $p = .05$.

Two statistically significant differences emerged from the posttest dependent measures. The wilderness backpack group differed in overall happiness from the other two groups, $F(4, 130) = 3.07$, $p < .01$.² Tests of simple main effects revealed that the groups differed at the follow up, $F(2, 36) = 4.16$, $p < .02$, but not at the pretest, $F(1, 40) = 1.95$, $p < .17$, nor the immediate posttest, $F(1, 40) = .81$, $p < .37$. No differences were found between groups on any of the ZIPERS factors.

Table 3 reveals that for the proofreading results, as with the overall happiness results, the groups were roughly equivalent prior to treatment, $F(2, 65) = 1.78$, $p < .17$. However, a Time $\times$ Group interaction occurred, with the backpacking group improving in proofreading performance, $F(2, 65) = 2.39$, $p < .09$, whereas the other two groups declined.

The initial study makes several contributions. First, it substantiates findings from previous studies by showing that a prolonged wilderness experience has restorative effects. These effects are manifest in some self-reports of affect. They also appear, to our knowledge for the first time, in a behavioral measure of restoration from mental fatigue. The multimethod convergence is important given the monomethod approach that has been predominant in this area of inquiry.

Second, the results are probably not explainable by self-selection rival hypotheses, given the controls built into the design (all physically fit backpackers, vacation and nonvacation control groups), similarity of groups prior to treatment, use of covariates, and use of a longitudinal design.

Third, the data suggest, contrary to expectations, that a wilderness experience may not immediately lead to greater positive affect, and may even slightly depress mood (see Table 2). On reflection these results are not surprising. The immediate emotional response on returning to one's usual setting may include negative feelings from seeing too many people and too much pollution, and from an awareness that one must return to the hectic pace and demands of one's daily life.

Fourth, the overall happiness data from the long-term follow-up suggest that some of the initially depressed reactions at the end of the backpack trip not only dissipate but actually reverse over time. One intriguing implication of this finding, should it prove robust, is that restorative experiences may also have proactive effects, preparing people to better cope with the stress and strain of daily life.

STUDY 2

A second study was designed to extend some of the exploratory findings of Study 1 under conditions of greater methodological rigor. A true experimental design was employed, with random assignment of college students to either a nature walk, a walk in an urban setting, or a relaxation condition. Because random assignment substantially reduces the plausibility of self-selection alternative hypotheses, the sample was not limited to experienced, physically fit backpackers. Although Study 1 improved on prior research in terms of internal validity it did so at some cost for external validity, given the restricted population sampling.

In addition to the stronger internal validity and greater generalizability afforded by its design, a third advantage of Study 2 is that it allows us to examine whether a walk in a park setting provides greater benefits than does simply relaxing in a comfortable chair. This is similar in concept to the nonwilderness vacation condition in Study 1.

An important limitation of Study 1 and the prior research on nature as a restorative environment has been the absence of

direct assessment of underlying theoretical constructs believed to be operating. An important additional objective of Study 2 was to provide a more theoretically relevant examination of natural settings as restorative environments. To accomplish this, three key components were built into the design of Study 2.

First, a scale was developed to measure restorative aspects of person-setting interactions. The scale was based on the S. Kaplan and Talbot (1983; see also R. Kaplan & S. Kaplan, 1989) model of restorative environments. Items were constructed to measure individual perceptions of fascination, being away, coherence (extent), and compatibility.

Second, cognitive fatigue was induced in all participants just before their treatment experience (nature walk, urban walk, passive relaxation). If restoration is linked to the reduction of cognitive fatigue, then exhausting attentional capacity should aid detection of differential restoration. Standardized induction of mental fatigue should also help to ensure the uniformity of participants' general state prior to the treatment. Note that the induction of cognitive fatigue in Study 2 was unlike Ulrich's manipulations (1979; Ulrich & Simons, 1986) in that it was not designed to effect fear or threat. As noted earlier, Ulrich's experimental inductions of stress involved a college examination and an industrial accident film.

Third, physiological measurement was included in the methodology of Study 2. By including measures of both mental and physiological restoration, we can begin to distinguish between the relative contributions of mental fatigue and arousal models to our understanding of restorative environments.

Finally, because Ulrich's findings of greater restoration from natural scenes were obtained in laboratory settings with short-term simulations of different settings, it is important to examine their external validity with data obtained under more naturalistic conditions. Further, although overall happiness results from Study 1 are somewhat consistent with Ulrich's findings, the lack of differences in the ZIPERS data raises questions about the reliability of Ulrich's (1979, 1981) ZIPERS findings. These questions are also considered in the present study.

Method

Participants. All participants in Study 2 were college students who participated in exchange for course credit in introductory social science courses. Seventeen men and 17 women ($M = 20$ years) were randomly assigned to each of three conditions: natural environment experience, urban environment experience, and passive relaxation experience.

Dependent Measures. As in Study 1, the Zuckerman Inventory of Personal Reactions (ZIPERS) and Overall Happiness scale were used to measure emotional restoration. The same proofreading materials were used to measure mental restoration. Blood pressure and pulse were measured through an automatically inflated cuff on the dominant arm with a Critikon Dinamap Vital Signs Monitor (Model 1846P). Readings were taken every 4 minutes while participants sat comfortably. Skin conductance was monitored with a Psyonics AGSC device through Beckman 16 mm silver-silver chloride electrodes fixed on the palm of the nondominant hand. Johnson and Johnson K-Y surgical jelly was used as a conductive medium.

A measure of perceived restorative quality was developed in accordance with the Kaplans' (S. Kaplan & Talbot, 1983) theoretical model. Participants made their evaluations on 5-point Likert-type scales of being away (e.g., I felt removed from my everyday environment), fascination (e.g., My attention was drawn to many interesting things during my time in the setting), coherence/extent (e.g., I gained a sense of the way the elements of the setting fit together), and compatibility (e.g., I saw that some of the things I like to do I could do there). The internal consistency of the 20-item scale was high (Cronbach's $\alpha = 0.92$).

Procedure. In an initial baseline session, resting measures of physiology were taken for a 20-minute period following administration of the ZIPERS and OHS. The proofreading measure was completed after the physiological measurement.

The experimental session was held on a different day in one of three settings: (a) a park with a stream and associated riparian habitat, chaparral flora, some ornamental vegetation,

and various rock outcroppings, located near a major metropolitan area (Santiago Oaks Regional Park, Orange, California); (b) a well-kept area with mixed residential and commercial uses near the center of a large, culturally diverse city (Santa Ana, California); or (c) the same laboratory setting in which the baseline session was conducted, located on the campus of the University of California, Irvine. The latter was furnished with a comfortable chair, magazines, radio, and a reading lamp. With the exception of treatments, the procedure at both field sites was carried out in a room similar in terms of furnishings to the laboratory setting.

On arrival at the assigned location, participants filled out a brief questionnaire on activities and mood for that day. Those in the field conditions also answered questions about their drive to the field site. Physiological measures were then taken for 20 minutes. This provided a second set of baseline data.

After providing initial measures, participants underwent 40 minutes of tasks designed to induce cognitive fatigue. Standardized presentations of instructions and task stimuli were made possible through the use of a prerecorded tape. Participants initially performed a Stroop task in which they were to read aloud the color of the ink various color words were written in (e.g., the word *red* written in green ink—correct response “green”; Stroop, 1935). Stimulus words were displayed on a chart with 20 rows and 5 columns. Cell coordinates were presented once every three seconds. The Stroop task was followed by a binary classification task in which numbers presented once every two seconds had to be classified as odd or even and high or low according to a given criterion. This sequence of tasks was repeated twice. To increase task motivation, participants were given a monetary incentive; a \$50 reward was offered for best performance.

Immediately following the tasks, each participant was either led individually on a prearranged walk in one of the field settings or sat quietly to read magazines for 40 minutes. At the outset of the walk the participant was asked to simply relax and be aware of the environment around him or her. The experimenter also

suggested, in a pleasant way, that conversation should be minimized.

Physiological measurement was resumed after completion of the walk/rest period. During an initial 8-minute period participants completed the perceived restorativeness scale, ZIPERS, and OHS, in that order. A final 20 minutes of physiological recording was then completed. This was followed by the proof-reading task. Further details on Study 2 methods can be found in Hartig (1990).

Results and Discussion

Even though participants were randomly assigned to conditions, a few differences among the groups were noted. For example, participants had to drive to get to the two field sites (21.5 miles on average to the park; 14.7 to the urban site), whereas the laboratory was located on campus. There was also some variation in lag time between baseline and experimental sessions. Finally, students in the relaxation group ($M = 3.0$ years) were slightly more senior in class rank than students in both of the other two groups ($M = 2.6$ years). To deal with these differences, an analysis of covariance strategy was employed when the difference between groups was also significantly related to the outcome variable under scrutiny. In Study 2 planned orthogonal comparisons of the posttest scores were employed throughout to test the hypothesis that the natural environment experience would be more restorative than the other two conditions. Posttest scores were used because the pretest scores were equivalent throughout.

As a manipulation check, summary scores on the perceived restorativeness scale were compared across the three groups, $F(2, 93) = 14.81, p < .001$ (with lag time between sessions and miles driven to site as covariates). Participants in the natural environment group ($M = 80.8$) experienced being away, fascination, coherence/extent, and compatibility to a greater degree than did participants in either the urban walk ($M = 66.2$) or relaxation conditions ($M = 57.8$).

TABLE 4
Group Means for Posttest Measures of Emotion

<i>Dependent Variable</i>	<i>Natural Environment</i>	<i>Urban Environment</i>	<i>Passive Relaxation</i>
Fear	1.49	1.45	1.39
Anger and aggression ^a	1.14	1.30	1.44
Sadness	1.17	1.35	1.29
Positive affect ^a	3.02	2.59	2.55
Attentiveness	3.14	3.21	3.09
Overall happiness ^a	81.4	74.4	71.8

NOTE: The Fear through Attentiveness measures were scaled from 1 = low to 5 = high. The Overall happiness measure was scored from 0 = Not at all to 100 = Very happy.

a. Natural environment group differs significantly from the other two groups.

Table 4 shows the means for the self-reports of mood and general affect taken after the treatment period. As can be seen in the table, the natural environment group had higher ratings of Overall Happiness, $t(90) = 1.68, p < .05$ and ZIPERS positive affect scores, $t(95) = 2.37, p < .01$. The park group also had reliably lower ZIPERS anger/aggression scores, $t(82.51) = 2.69, p < .005$.³ The overall pattern of data in Table 4 is consistent with the hypothesis that the natural setting experience would be more restorative than the urban walk or relaxation experiences.

The behavioral measure of recovery from mental fatigue also reflects the stronger positive impact of the natural setting ($M = 56.5$ % errors detected) in comparison to the urban ($M = 49.5$) and relaxation conditions ($M = 48$), $t(94) = 2.45, p < .01$. This finding is particularly important because it corroborates the behavioral finding from Study 1.

Group means for the physiological indices of restoration are presented in Table 5. As inspection of the table suggests and statistical analyses confirm, there were no significant differences among groups in blood pressure and heart rate. This lack of differences could have a procedural cause. Because physiology was not monitored while participants went through their treatment experiences, the final assessments of restoration were not obtained until approximately 50 minutes after completion of the tasks. Given that Ulrich and Simons (1986) showed

TABLE 5
Group Means for Posttest Physiological Measures

<i>Dependent Variable</i>	<i>Natural Environment</i>	<i>Urban Environment</i>	<i>Passive Relaxation</i>
Systolic blood pressure (mmHg)	118.0	116.7	117.1
Diastolic blood pressure (mmHg)	63.7	61.6	64.7
Heart rate (bpm)	65.5	64.7	66.9

return to baseline levels within 10 minutes following exposure to a stressful film, it is perhaps not surprising that our posttreatment means are not different from one another. A better alternative would have been to employ ambulatory monitoring during the treatment conditions. Unfortunately, the needed equipment was not available to us.

GENERAL DISCUSSION

Taken in combination, the results of the present studies provide relatively strong evidence that experiences in natural settings have restorative outcomes. Importantly, restoration was uncovered in both studies with a measure of attentional performance known to be a sensitive index of mental fatigue associated with noise, crowding, and other environmental demands (Cohen, 1980; Evans & Cohen, 1987). As indicated in the introduction, use of this measure in the present studies supplements prior research that has relied solely on self-report measures of restoration. Evidence of more complete recovery of attentional capability in both studies offers support for the claim that natural settings are restorative in part because they facilitate recovery from mental fatigue (S. Kaplan & Talbot, 1983; R. Kaplan & S. Kaplan, 1989).

The self-report data on affect and mood indicate a general pattern of restoration in natural settings consistent with the literature reviewed in the introduction. One anomaly bears further mention, however. As suggested by the Study 1 overall

happiness finding, the period immediately following a wilderness experience may be properly characterized as one of mixed feelings. Although people may feel rejuvenated, they may also experience negative thoughts and feelings about returning to their usual setting and situations. As indicated by other research as well (e.g., Clawson & Knetsch, 1966; Hammitt, 1980), future research on wilderness and other outdoor recreation experiences needs to more carefully monitor affect, cognitive capabilities, and physiological functioning at different phases in such experiences. We suspect, for example, that if data on mood and affect had been collected midway through the trip, the backpacking group might have expressed more positive feelings.

The importance of temporal factors in the wilderness experience is also manifest in the longer term positive emotional effect. That greater happiness prevailed over a 3-week period, although necessarily a tentative finding, certainly portends the value of more extended investigation of the residual impacts of nature on well-being. The data speak to the question of whether natural settings "inoculate" people against stress, attentional demands, and the like.

Despite the mixed wilderness experience findings, it appears that the more short-term exposure to nature either on site as in Study 2 or via simulation as in Ulrich's work does have an immediately salutary effect. It is noteworthy that the ZIPERS results from Study 2 approximate those of Ulrich (1979), who found a similar pairing of increased positive and decreased negative affect following exposure of participants to scenes of nature.

One of the characteristics of wilderness and other natural environments that is believed to be critical for their restorative qualities is being away from one's daily routine. Data from both Study 1 and Study 2 indicate that being away is not sufficient to produce restorative effects. In Study 1 backpackers on a non-wilderness vacation were less restored than backpackers on a wilderness trip. Moreover the former group did not differ from a third group of backpackers who stayed home. In Study 2 the

urban setting was of equally low familiarity to students as the park.

Another interesting implication of the findings is that restorative effects can be realized in a range of natural settings that includes urban parks and wilderness areas. Aside from the present data, R. Kaplan and S. Kaplan (1989) review several studies suggesting that gardening and other local contacts with nature can enable restorative experiences.

One issue not yet addressed here concerns the factors that make natural settings, whether wilderness or garden, restorative. In the introduction we reviewed the Kaplans' restorative environments theory, which emphasizes the importance of fascination (involuntary attention), being away, coherence/extent, and compatibility. We found in Study 2 that, as measured by our preliminary instrument, perceptions of restorativeness engendered by the natural setting were indeed stronger than those arising from interactions with the urban and laboratory settings.

To further investigate the role of the restorative factors, we also examined in Study 2 the correlations between outcomes and the summary score derived from the perceived restorativeness scale. Of interest, the summary score was moderately but reliably correlated with those outcomes that were significantly impacted by the treatment. The obtained correlations were with overall happiness, $r = .36, p < .001$; with ZIPERS anger/aggression, $r = -.33, p < .001$; with ZIPERS positive affect, $r = .42, p < .001$; and with proofreading performance, $r = .22, p < .03$). The restorativeness summary score was not reliably correlated with any other outcome measures in Study 2.

Although the findings associated with the perceived restorativeness scale are promising, it must be noted that the scale is still in the initial stage of development. To date, only 20 items have been tested with a modest sized sample and with respect to a very limited range of settings. With development of a more psychometrically sound instrument, it may be possible to more directly investigate several important research issues. For example, a better instrument may enable examination of the

contributions to restoration from specific aspects of natural settings. One might further explore the importance of water, greenery, vistas, mystery, and such, as well as the linkages these features have with the different restorative factors described by R. Kaplan and S. Kaplan (1989). One might also try to compare the relative importance of these emergent properties themselves.

Our empirical studies raise the possibility that carefully constructed therapeutic environments may facilitate recovery from mental fatigue induced by certain of the demands of modern life. Moreover, we have raised the intriguing but essentially unexplored issue of inoculation. Can environments be configured so that people can proactively withstand the demands of contemporary society? Finally, our research adds weight to the belief that, as well as sustaining life, natural environments help foster, to paraphrase John Muir, inner peace and a renewal of mental energy.

NOTES

1. Additional measures were collected but are not reported here. One additional measure of affect is not included because the results are redundant with those presented. A second measure of mental restoration, attempts at solution of insoluble puzzles, is not included for several reasons. This measure followed the proofreading measure and thus its status as an aftereffect index is unclear. In addition, *F* values were less than one and very large standard deviations occurred. Finally, a measure of spirituality was dropped because there was little variance and the validity of the measure is not well developed.

2. Degrees of freedom vary because of missing data and use of covariates.

3. This comparison carried out with Welch's procedure for nonpairwise comparisons (Welch, 1947; cited in Kirk, 1982). The procedure is appropriate when variances are heterogeneous and groups sizes are unequal.

REFERENCES

- Campbell, A., Converse, P. E., & Rodgers, W.L. (1976). *The quality of American life*. New York: Russell Sage.

- Clawson, M., & Knetsch, J. L. (1966). *Economics of outdoor recreation*. Baltimore, MD: Johns Hopkins Press.
- Cohen, S. (1978). Environmental load and the allocation of attention. In A. Baum, J. E. Singer, & S. Valins (Eds.), *Advances in environmental psychology: Vol. 1. The urban environment* (pp. 1-29). Hillsdale, NJ: Lawrence Erlbaum.
- Cohen, S. (1980). Aftereffects of stress on human performance and social behavior: A review of theory and research. *Psychological Bulletin*, 88(1), 82-108.
- Cohen, S., & Spacapan, S. (1978). The aftereffects of stress: An attentional interpretation. *Environmental Psychology and Nonverbal Behavior*, 3, 43-57.
- Driver, B. L., & Knopf, R. C. (1976). Temporary escape: One product of sport fisheries management. *Fisheries*, 1, 21-29.
- Driver, B. L., Nash, R., & Haas, G. (1987). Wilderness benefits: A state-of-knowledge review. In R. C. Lucas (Ed.), *Proceedings—National Wilderness Research Conference: Issues, state-of-knowledge, future directions* (pp. 294-319). Ogden, Utah: USDA Forest Service Intermountain Research Station.
- Evans, G. W., & Cohen, S. (1987). Environmental stress. In D. Stokols & I. Altman (Eds.), *Handbook of environmental psychology: Vol. 1* (pp. 571-610). New York: Wiley.
- Ewert, A. (1988). Research in outdoor pursuits: Overview and analysis. In J. Cederquist (Ed.), *Exploring pathways to adventure: Proceedings of the 1986 Conference on Outdoor Recreation* (pp. 103-120). Salt Lake City: Outdoor Program, University of Utah.
- Gibson, P. M. (1979). Therapeutic aspects of wilderness programs: A comprehensive literature review. *Therapeutic Recreation Journal*, 13(2), 21-33.
- Glass, D. C., & Singer, J. E. (1972). *Urban stress: Experiments on noise and social stressors*. New York: Academic Press.
- Hammitt, W. E. (1980). Outdoor recreation: Is it a multi-phase experience? *Journal of Leisure Research*, 12(2), 107-115.
- Hartig, T. (1990). *Testing the theory of restorative environments*. Unpublished master's thesis, University of California, Irvine.
- Hendee, J. C., Catton, W. R., Jr., Marlow, L. D., & Brockman, C. F. (1968). *Wilderness users in the Pacific Northwest: Their characteristics, values, and management preferences* (USDA Forest Service Research Paper PNW-61). Portland, OR: USDA Forest Service Pacific Northwest Forest Experiment Station.
- James, W. (1892). *Psychology: The briefer course*. New York: Holt.
- Kaplan, R. (1974). Some psychological benefits of an outdoor challenge program. *Environment and Behavior*, 6(1), 101-116.
- Kaplan, R. (1977). Summer outdoor programs: Their participants and their effects. In *Children, nature, and the urban environment: Proceedings of a symposium fair* (USDA Forest Service General Technical Report NE-30, pp. 175-179). Upper Darby, PA: USDA Forest Service Northeastern Forest Experiment Station.
- Kaplan, R., & Kaplan, S. (1989). *The experience of nature: A psychological perspective*. Cambridge: Cambridge University Press.
- Kaplan, S. (1978). Attention and fascination: The search for cognitive clarity. In S. Kaplan & R. Kaplan (Eds.), *Humanscape: Environments for people* (pp. 84-90). Belmont, CA: Duxbury (Ann Arbor, MI: Ulrich's Books, 1982).
- Kaplan, S. (1983). A model of person-environment compatibility. *Environment and Behavior*, 15(3), 311-332.

- Kaplan, S. (1987). Mental fatigue and the designed environment. In J. Harvey & D. Henning (Eds.), *Public Environments* (pp. 55-60). Washington, DC: Environmental Design Research Association.
- Kaplan, S., & Talbot, J. F. (1983). Psychological benefits of a wilderness experience. In I. Altman & J. F. Wohlwill (Eds.), *Human behavior and environment: Vol. 6. Behavior and the natural environment* (pp. 163-203). New York: Plenum.
- Kirk, R. E. (1982). *Experimental design: Procedures for the behavioral sciences*. Belmont, CA: Brooks/Cole.
- Knopf, R. C. (1983). Recreational needs and behavior in natural settings. In I. Altman & J. F. Wohlwill (Eds.), *Human behavior and environment: Vol. 6. Behavior and the natural environment* (pp. 205-240). New York: Plenum.
- Knopf, R. C. (1987). Human behavior, cognition, and affect in the natural environment. In D. Stokols & I. Altman (Eds.), *Handbook of environmental psychology: Vol. 1* (pp. 783-825). New York: Wiley.
- Levitt, L. (1989). Therapeutic value of wilderness. In *Wilderness Benchmark 1988: Proceedings of the National Wilderness Colloquium* (USDA Forest Service General Technical Report SE-51; pp. 156-168). Asheville, NC: USDA Forest Service Southeastern Forest Experiment Station.
- Mang, M. (1984). *The restorative effects of wilderness backpacking*. Unpublished doctoral dissertation, University of California, Irvine.
- Myers, J. L. (1972). *Fundamentals of experimental design* (2nd ed.). Boston: Allyn & Bacon.
- Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology*, 18(6), 643-662.
- Ulrich, R. S. (1979). Visual landscapes and psychological well-being. *Landscape Research*, 4(1), 17-23.
- Ulrich, R. S. (1981). Natural versus urban scenes: Some psychophysiological effects. *Environment and Behavior*, 13(5), 523-556.
- Ulrich, R. S. (1983). Aesthetic and affective response to natural environment. In I. Altman & J. F. Wohlwill (Eds.), *Human behavior and environment: Vol. 6. Behavior and the natural environment* (pp. 85-125). New York: Plenum.
- Ulrich, R. (1984). View through a window may influence recovery from surgery. *Science*, 224, 420-421.
- Ulrich, R. S., & Simons, R. F. (1986). Recovery from stress during exposure to everyday outdoor environments. In Barnes, R., Zimring, C., & Wineman, J. (Eds.) *The costs of not knowing* (pp. 115-122). Washington, DC: Environmental Design Research Association.
- Welch, B. L. (1947). The generalization of Student's problem when several different population variances are involved. *Biometrika*, 36, 293-296.
- Zajonc, R. B. (1980). Feeling and thinking: Preferences need no inferences. *American Psychologist*, 35, 151-175.
- Zuckerman, M. (1977). Development of a situation-specific trait-state test for the prediction and measurement of affective responses. *Journal of Consulting and Clinical Psychology*, 45, 513-523.